Representing images: pixels, resolution, bitmap and vector

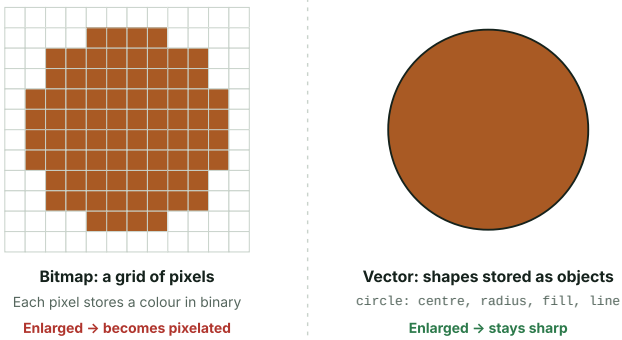
L03 FACT FILE

Specification: students should be able to

- Demonstrate understanding of how pixels are used in image representation
- Demonstrate understanding of how image resolution affects file size
- Describe how vector-based graphics and bitmap graphics are stored
- Describe the difference between vector-based and bitmap graphics
- Calculate the file size of a bitmap image from its resolution and colour depth

CCEA fact file: Digital Data 02: Representing Images

Key words: pixel, bitmap, vector, resolution, colour depth, file size, pixelation, metadata, coordinates, ppi / dpi, object, properties, scaling, SVG, 2-bit colour



Bitmap image as a grid of pixels compared with a vector circle stored as an object

Pixels and bitmap images

A pixel, short for picture element, is the smallest part of a digital image that can be shown and edited on a screen. A bitmap image is a grid of pixels, and each pixel can be a different colour. When all the pixels are seen together they make the complete picture. Each pixel's colour is stored as a binary code, and its place in the grid is given by coordinates, like (x, y) on a graph. When you take a photo on your phone or scan a page, you create a bitmap. To save a bitmap and display it again, the computer stores the colour code of every pixel plus metadata, which is data about the image such as its width, height and colour depth. Without the metadata, the computer would have a long list of colour codes but would not know where each row of the picture ends.

Colour depth

Colour depth is how many bits are given to each pixel to record its colour. The more bits per pixel, the more different colours the image can contain. One bit gives two possible codes, 0 and 1, so only two colours such as black and white. Two bits give four codes (00, 01, 10 and 11), so four colours. Every extra bit doubles the number of colours, so 8 bits give 256 colours and 24 bits give about 16.7 million colours. Most photographs use 24-bit colour, with 8 bits each for the amounts of red, green and blue. To work out how many bits you need, find the smallest power of 2 that is at least the number of colours. For example, an image with 5 colours needs 3 bits, because 2 bits only give 4 codes. More colours make an image more realistic but increase the file size.

Resolution and file size

Resolution describes how many pixels an image contains. It is usually written as two numbers, such as 1280 × 720: the first is the number of pixels across and the second is the number of pixels down. It can also be measured as pixels per inch (ppi) on a screen, or dots per inch (dpi) when printing. The higher the resolution, the more pixels there are, so the image is sharper and shows more detail. However, every pixel must be stored, so higher resolution means a larger file that takes more storage space and longer to load or upload. To estimate the size of the image data, multiply width by height by colour depth to get bits, then divide by 8 for bytes and by 1024 for kilobytes. A 10 × 10 image with 4 colours uses 10 × 10 × 2 = 200 bits, which is 25 bytes.

Vector graphics

Vector graphics are not made from a grid of pixels. Instead, the file stores a list of the objects that make up the picture, such as lines, curves, rectangles and circles, together with their properties. Typical properties are the position of the object, its size, its line colour, line thickness and fill colour. Because these objects are described mathematically, the computer recalculates and redraws them whenever the image is resized. This means a vector graphic can be enlarged as much as you like without losing quality or becoming pixelated. Each object can be selected and edited on its own, for example moving a shape or changing its colour. Vector files are usually smaller than bitmaps because they store a description of each shape rather than every pixel. They are ideal for logos, diagrams, maps and fonts, but not for photographs, which have too much colour detail. SVG is a common vector format.

Comparing bitmap and vector

The key difference is how each type is stored. A bitmap stores the colour of every pixel in a grid, while a vector stores objects and their properties. This leads to other differences. When a bitmap is enlarged, its pixels get bigger and the image becomes blocky, which is called pixelation; a vector is recalculated, so it stays sharp at any size. Bitmap file sizes grow with resolution and colour depth and can be very large, while vector files are usually smaller. A bitmap is edited pixel by pixel, whereas a vector is edited one whole object at a time. Bitmaps are best for photographs and scanned images where colours change gradually, and vectors are best for images built from shapes, such as a school crest that must look sharp both on a small badge and on a large sign at the school gate.

Key terms from the mark schemes

Pixel
The smallest element of a digital picture that a screen can show or an editor can change. Every pixel holds one colour, recorded as a binary code.
`Credited: picture element`

Bitmap graphic
An image made of a grid of pixels, each of which can be a different colour. The colour code of every pixel is stored, plus metadata such as the dimensions. Pixelation occurs when it is enlarged. Example file type: JPEG.
`Credited: raster image`

Vector graphic
An image stored as a list of objects (lines, curves and shapes) with properties such as position, size, line colour and fill colour. Can be enlarged without loss of quality. Example file type: SVG.
`Credited: vector-based graphic`

Resolution
The number of pixels in an image (width × height) or per inch. More pixels give a sharper image but a bigger file size.
`Credited: image resolution`

Colour depth
The number of bits used to store the colour of each pixel. n bits give 2^n colours.
`Credited: bits per pixel`

Pixelation
The blocky appearance of a bitmap image when it is enlarged too much, because the individual pixels become visible.

PPI / DPI
Pixels per inch (screen) / dots per inch (printing): measures of resolution.

Metadata
Data about the data, e.g. the width, height and colour depth of an image, stored so that the image can be rebuilt for display.

Exam focus

Bitmap and vector graphics appear regularly. 2023 Q2(a) and 2025 Q2(a) asked pupils to describe differences or decide which statements apply to bitmap or vector graphics. For a 'describe two differences' question, each mark normally needs a comparison, such as 'a bitmap is stored as a grid of pixels whereas a vector is stored as objects and their properties'. Resolution and file size were tested in 2019 Q4(a) and 2023 Q2(b): you must say that more pixels give a sharper image but a larger file. 2019 Q4(a)(iii) asked for units of resolution, so learn ppi and dpi. Defining a pixel needs 'smallest unit of a digital image'; simply saying 'a dot' is too vague. In file size calculations, write the formula, show each step and give units. Common mistakes are forgetting to divide by 8 for bytes and using the number of colours instead of the number of bits.

Unit 1 2025 Q2	[6]	Bitmap vs vector; sound file size calculation
Unit 1 2023 Q2	[10]	Bitmap vs vector, resolution, buffering, sample rate, ADC, compression
Unit 1 2019 Q4(a)	[6]	Image file type, compression, resolution, ppi/dpi

? Check yourself

1. Define the following terms: (a) pixel (b) image resolution. [2]
2. Explain how increasing the resolution of an image affects its quality and its file size. [3]
3. Describe how a vector graphic is stored. [2]

Answers: 1. (a) The smallest unit / element of a digital image that can be shown on screen and edited (1). (b) The number of pixels in an image, given as width × height / pixels per inch (1). 2. More pixels are used (1); so the image is sharper / shows more detail / higher quality (1); but more data must be stored so the file size increases (1). 3. Stored as a list of objects / shapes such as lines, curves and circles (1); with their properties, e.g. position / coordinates, size, line colour, fill colour, line thickness (1).